Meeting 2 of 7 · September 29, 2026

Retrospective reconstruction · prepared October 2026

RESEARCH DESIGN

Designing the study: model, mitigation and evaluation

Research design: scope, variational classifier template, zero-noise extrapolation, evaluation plan

Reconstructed in Oct 2026 from the project record, which begins Oct 6. It preserves this stage as a generic QML design (Iris, QM9, WDBC) with a mitigate-or-not decision. The fraud application came on Oct 6.

Mayank Kumar Pokhriyal · Tushar Poonia
M.S. in Artificial Intelligence capstone · Katz School of Science and Health, Yeshiva University
Retrospective deck · proposed split: Tushar (slides 1–3) · Mayank (slides 4–5) · about 5 minutes



Scope as recorded before October 6

Recorded for this stage

Benchmarks
Iris and WDBC (classification), QM9 (molecular properties); Qiskit Aer

Decision
mitigate or not (ZNE / readout mitigation)

Approach
Q-TRACE: benefit-minus-cost rule vs Never / Always / Fixed

Adopted later, on Oct 6, 2026

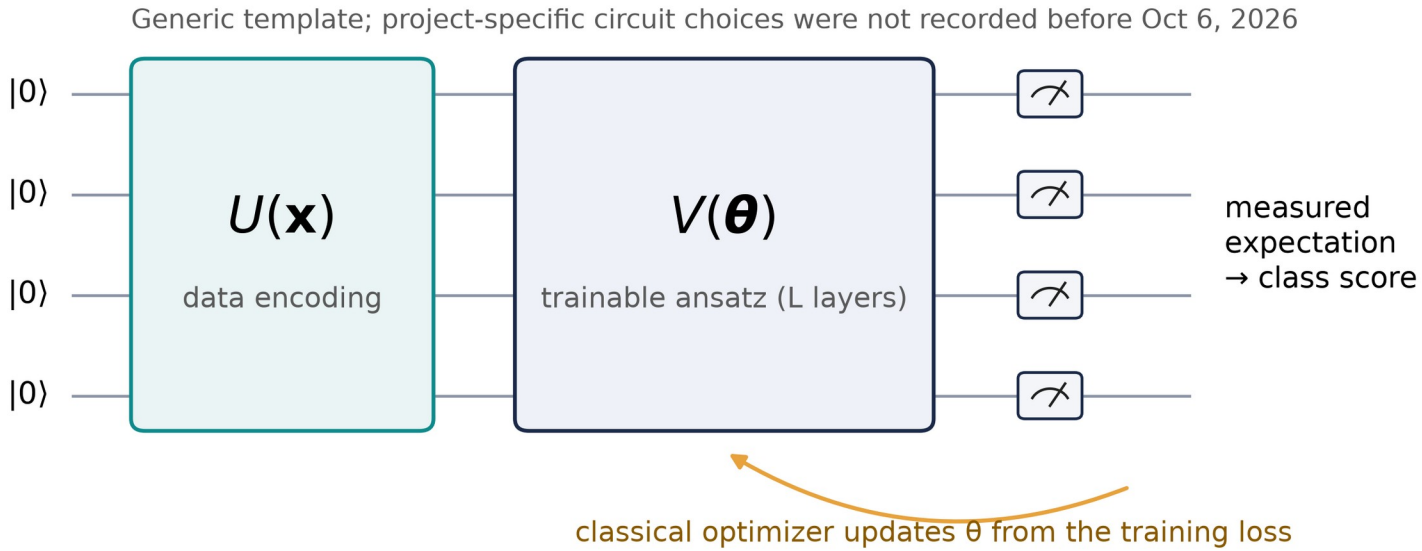
- Application: credit-card fraud detection
- Several mitigation actions instead of a binary choice
- Label-free decision inputs only (A01, A12)
- Pre-registered, gated phases

Source: docs/DECISIONS.md (Q1, A01, A12), docs/PROJECT_SPEC.md

We present this stage as recorded; the fraud application is a later decision, not part of it.

Source: docs/DECISIONS.md (Q1, answered 2026-10-06), docs/PROJECT_SPEC.md §2, docs/research/deep-research-report.md

The variational quantum classifier template



Why it matters here

- The model's output is a measured expectation value
- Noise distorts exactly that quantity
- Mitigation tries to repair it, at a cost

A standard, well-studied model template, so the study is about the mitigation decision, not a new classifier.

Source: Havlíček et al. 2019 (Nature 567:209) · Schuld et al. 2020 (PRA 101:032308). Generic template.

Mitigation strategy: zero-noise extrapolation

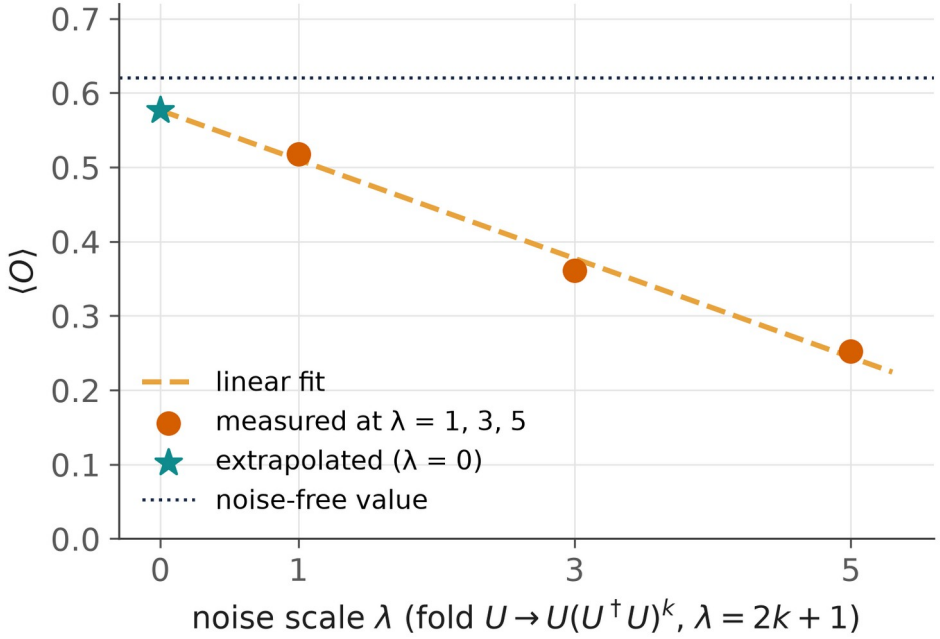
1. Amplify
fold: U → U U† U (same ideal result, more noise)

2. Measure
at several noise levels

3. Extrapolate
fit the trend back to zero noise

The cost behind the decision

- Every extra noise level is another circuit to run
- More shots and more two-qubit-gate exposure
- So: is the expected improvement worth it here?



Illustrative values (not project data).

ZNE can remove noise-induced bias, but always at a resource cost.

Source: Temme et al. 2017 (PRL 119:180509) · Giurgica-Tiron et al. 2020 (IEEE QCE). Curve illustrative.

How the decision would be evaluated

Settings
benchmark × noise level

→

Run twice
with and without mitigation

→

Record
quality change + extra resources

→

Compare policies
Never · Always · Fixed · Q-TRACE

Never
cheapest; leaves noise-induced bias

Always
pays the full cost every time

Fixed rule
one preset threshold

Q-TRACE
mitigate when expected benefit exceeds cost

Not recorded at this stage: exact metrics, noise models and thresholds (fixed from Oct 6). The example input then, "current accuracy", needs labels and was removed on Oct 6 (A12).

Evaluate Q-TRACE against never, always and a fixed rule.

Source: docs/DECISIONS.md (Q1, A12), docs/research/deep-research-report.md (pre-October deck); wording is ours.